Homework 8

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1

Staring at the table, the outlier stock is stock C — the others have expected return increase by 0.07 each time their β increases by 0.1. The expected return of stock C, based on the slope of the SML and the given values for stock D, is $0.068 + 3 \times 0.07 = 0.089$, so the stock's alpha is $\boxed{\alpha_C = 0.103 - 0.089 = 0.014}$ since this gives the extra return above the expected value from the SML.

2

The two non-systemic biases are familiarity bias and overconfidence bias — these are evidenced more for individual investors than for the broader population as a whole.

3

Under a multi-factor model, we write the expected return as

$$\mathbb{E}[R] = r_f + \sum \beta_S^{F_n} \left(\mathbb{E}[R_{F_n}] - r_f\right)$$

We are given $\mathbb{E}[R_A] = 0.25$ and $\mathbb{E}[R_B] = 0.2$, $\beta_S^A = 0.1$ and $\beta_S^B = 0.15$, and $r_f = 0.05$. Thus, under the two-factor model, we can write

$$\mathbb{E}[R_S] = 0.05 + 0.1 \times (0.25 - 0.05) + 0.15 \times (0.2 - 0.05) = \boxed{0.0925 = \mathbb{E}[R_S]}$$